

## ASTR211 Data Reduction Log

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Date: 8/824

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Dataset Option: Jupiter

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Process your data using your code from Lab 3. Record your process below, in particular including:

### A summary of your data

How many of each type of image, in which filter, at which exposure time.

What sets will you use to create each of your master darks and master flat.

- 33 darks exptimes between 0.15 and 300
- 83 flats exptimes between 3 and 30
- 76 raws exptimes between 0.01 and 120 filters are in methane, OIII, SII and Halpha

### Working out a problem

Expect to encounter problems. For each problem document:

1. What is the problem?
2. How did you find it?
3. How does this affect the raw, calibration and/or science images?
4. How did/would you solve it?

Record below the process of reducing your data.

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## Pulling in the fits files from Macdiarmid

- First problem, it seems some of the fits files don't have an imagetype specification.
- When I go to sort by dark/flat/raw I get to a file that doesn't have an image
- Solved by first checking for IMAGETYP key in header dictionary, if not present, assume file is a dark
- It has become obvious that one of the darks is missing the imgtype item.

## File Structuring

- For some reason the darks all have wildly varying exposure times. Between 0.15 and 300 seconds
- The same is true for both the flats and the lights. Most of the light exposure are short. There are a couple that are 2 minutes long which seems like it would be too long to get anything useful. I am going to sort the images into a dictionary based on exposure time to see what I am working with.

## Image Processing

- I have been advised to figure out which filter to use first before worrying about image type
- I will plot one of each filter type to figure out which picks up the spot the best
- After some filter dabbling it is clear that only the methane narrow band filter will be of use
- I will now discard all non methane images and the two 120 second exposures
- After displaying one of each exposure time i will also discard the 0.1s and 0.05s exposures as they are underexposed and will be a pain to make darks for.
- I am now sorting the flats by filter type to acquire all the flats with the methane filter
- After making a 3s and 4s master darks and making a combined master flat i thought I had an issue with my master flat but it turns out its just the methane filter
- No problems with getting out the science images

## Science

- First problem trying observe the rotation is that jupiter moves within the Frame so it isn't super easy to see the spot move.
- This problem isn't really a problem since we just need to see how far it moves on jupiter.
- Couldn't quite get the distance the spot moves.